

AISSAOUI Ismail

AI & Computer Vision Engineer | PhD Candidate

Email: ismail.aissaoui.pro@gmail.com Phone: +213 660 70 77 96

Portfolio: ismail-aissaoui.vercel.app Location: Algeria (Relocation Ready)

LinkedIn: linkedin.com/in/aissaoui-ismail-6a77a92a7 GitHub: github.com/i-aissaoui

PROFILE

AI & Computer Vision Engineer specialized in **Facial Recognition**, **Video Representation Learning**, and **Temporal Dynamics**. Expert in building robust vision pipelines with PyTorch and OpenCV. Proven ability to implement state-of-the-art architectures (ViT, Mamba-SSM) for complex sequential data. Deeply committed to the application of AI in Healthcare and Social Science (ASD Support).

EDUCATION

National Higher School of Computer Science (ESI-SBA) 2021–2026
Engineering & Master's Degree in Artificial Intelligence
Master's Thesis: "Modeling Temporal Dynamics for Industrial Systems using SSMs and xLSTM"
Focus: Sequential Representation Learning, Domain Generalization, Real-time Inference.

TECHNICAL SKILLS

Computer Vision: Facial Expression Recognition (FER), Image Processing, Object Detection (YOLOv8), Tracking (ByteTrack), Patch-based Attention (ViT).
AI & ML: PyTorch, TensorFlow, OpenCV, State-Space Models (Mamba), Transformer Architectures, Domain Generalization, Self-Supervised Learning.
Tools & Infrastructure: Git, Linux, Docker, CUDA, C++ JIT, InfluxDB, PostgreSQL.
Web: FastAPI, Next.js, WebSockets (for real-time vision streaming).

PROJECT HIGHLIGHTS

Recognizini — Real-Time Facial Recognition System 2025
LIRIS Focus: FER, Embeddings, Real-time Processing

- Developed an end-to-end facial recognition application featuring real-time identification and an active feedback loop for incremental embedding updates.
- Optimized facial feature extraction pipelines to handle varying head poses and illumination conditions.

Vision Transformer (ViT) Implementation 2025
LIRIS Focus: Foundation Models, Attention Mechanisms

- Implemented the seminal Vision Transformer architecture from scratch in PyTorch, focusing on linear projection of image patches and multi-head self-attention.

Industrial Temporal Modeling (Master's Thesis) 2025–2026
LIRIS Focus: Video Representation, Synchrony, SSMs

- Applied **Mamba-SSM** and **xLSTM** to model long-range temporal dependencies in high-frequency sensor data.
- Engineered models for **Domain Generalization**, enabling transfer between different fleet units with 98% reliability.

Real-time Safety & Traffic Analytics (YOLOv8) 2025
LIRIS Focus: Monitoring, unconstrained scenes, OpenCV

- Built real-time monitoring systems for PPE compliance and traffic flow, utilizing ByteTrack for consistent object tracking across frames.

LANGUAGES

English: Fluent (C1) | French: Fluent (C1) | Arabic: Native